

MOSSES

Mosses are non-flowering plants that typically form mats or cushion-shaped clumps. Despite their small size, they are remarkably resilient plants. They are able to grow in a wide range of habitats, from woodland to deserts, and are found on every continent, including Antarctica.

DIVISION	BRYOPHYTA
CLASSES	8
ORDERS	30
FAMILIES	110
SPECIES	About 10,000



In the far north sphagnum mosses form raised bogs – here interspersed with grey patches of reindeer moss lichen.

Mosses have thin leaves that usually spiral around slender, wiry stems, and they reproduce by scattering spores. Like liverworts, they need moist conditions to grow. They can be extremely abundant in habitats that are always damp, and some species – particularly sphagnum mosses – form extensive soggy blankets, dominating the ground in cold parts of the world. Others, however, are able to remain dormant during droughts. They look grey and lifeless, but become green again within minutes if it rains.

Mosses have a two-stage life-cycle – a feature shared by all other plants. In mosses, the dominant stage is the gametophyte, which produces male and female cells. After the female cells are fertilized, they produce the spore-forming stages or sporophytes, which remain attached to the parent plant. Most moss sporophytes have a capsule on the end of a long stalk. Once ripe – a process that can take many months – the capsule splits open, releasing up to 50 million spores into the air.

Because moss spores are so small and light, they are carried great distances by the slightest breeze. This makes mosses very good at dispersing and enables them to colonize all kinds of micro-environments, from crevices in the bark of trees to damp walls and roofs.

RELATIVES AND LOOK-ALIKES

The division name Bryophyta is sometimes used to encompass the liverworts and hornworts (see pp.110–111), but these are now regarded as separate, but closely related, taxonomic divisions. Clubmosses are more complex than true mosses, and belong in the division Lycophyta (see p.111). Other moss look-alikes include reindeer moss – actually a form of lichen – and Spanish moss, a flowering plant in the bromeliad family, that grows festooned on trees.

✓ DEEP COVER

The cool, moist climate in New Zealand’s Fjordland National Park enables a wide variety of mosses and liverworts to thrive.

